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BMAT202P – Probability and Statistics Lab

Digital Assessment 5

Two Sample Test and Chi – square Test

1. A manufacturer wants to know whether two production lines have the same average output. From Line A, a sample of 50 items has a mean output of 125 units, and the population standard deviation is 15. From Line B, a sample of 45 items has a mean output of 119 units, and the population standard deviation is 14. Test at the 1% level of significance whether Line A has a higher mean output than Line B.

Code- # Given data

```
x1 <- 125 # mean of Line A
```

```
x2 <- 119 # mean of Line B
```

```
s1 <- 15 # std dev of Line A
```

```
s2 <- 14 # std dev of Line B
```

```
n1 <- 50
```

```
n2 <- 45
```

Calculate Z statistic

```
z <- (x1 - x2) / sqrt((s1^2 / n1) + (s2^2 / n2))
```

Significance level

```
alpha <- 0.01
```

Critical value (right-tailed test)

```
z_critical <- qnorm(1 - alpha)
```

p-value

```
p_value <- 1 - pnorm(z)
```

Output results

```
cat("Z statistic =", z, "\n")
```

```
cat("Critical value =", z_critical, "\n")
```

```
cat("p-value =", p_value, "\n")
```

Decision

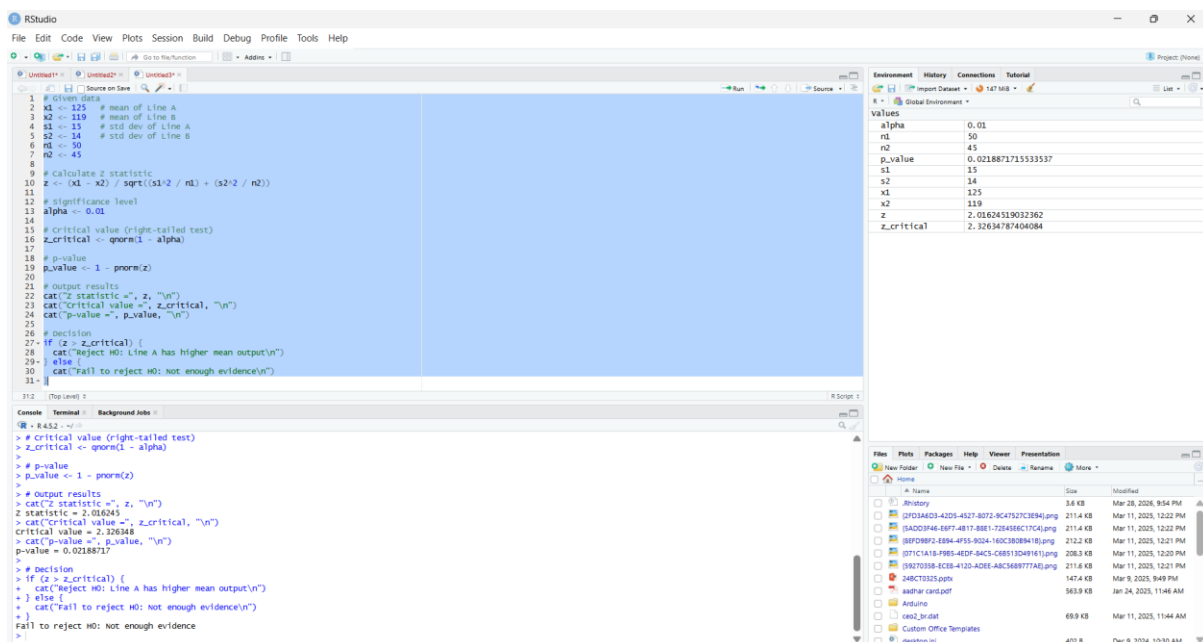
```
if (z > z_critical) {
```

```
  cat("Reject H0: Line A has higher mean output\n")
```

```
} else {
```

```
  cat("Fail to reject H0: Not enough evidence\n")
```

```
}
```



2. A hospital compares the recovery success rate of two treatments. In Treatment A, 86 out of 150 patients recovered successfully, while in Treatment B, 70 out of 160 patients recovered successfully. Test at the 5% level of significance whether the success proportions are different.

Code-

Given data

```
x1 <- 86
```

```
n1 <- 150
```

```

x2 <- 70
n2 <- 160

# Sample proportions
p1 <- x1 / n1
p2 <- x2 / n2

# Pooled proportion
p <- (x1 + x2) / (n1 + n2)

# Z statistic
z <- (p1 - p2) / sqrt(p * (1 - p) * ((1/n1) + (1/n2)))

# Significance level
alpha <- 0.05

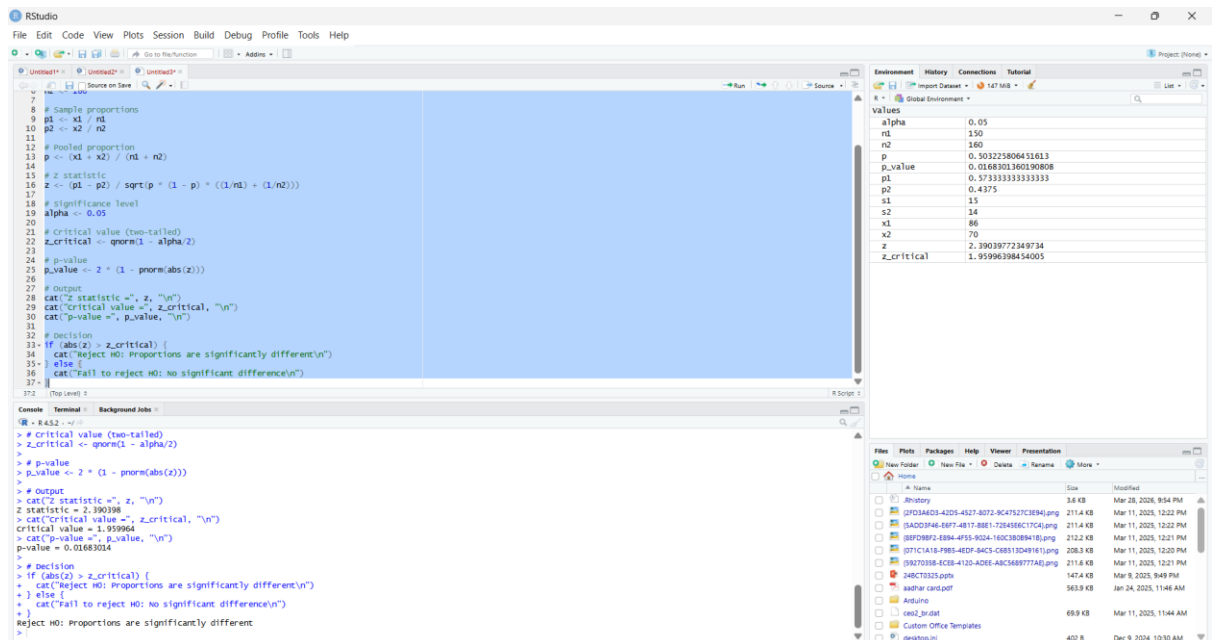
# Critical value (two-tailed)
z_critical <- qnorm(1 - alpha/2)

# p-value
p_value <- 2 * (1 - pnorm(abs(z)))

# Output
cat("Z statistic =", z, "\n")
cat("Critical value =", z_critical, "\n")
cat("p-value =", p_value, "\n")

# Decision
if (abs(z) > z_critical) {
  cat("Reject H0: Proportions are significantly different\n")
} else {
  cat("Fail to reject H0: No significant difference\n")
}

```



3 A researcher compares reaction times from two training programs. Program A is applied to 8 participants, and the times are 2.8, 3.1, 2.9, 3.0, 3.2, 2.7, 3.0, and 2.8 seconds. Program B is applied to another 8 participants, and the times are 3.4, 3.7, 3.5, 3.8, 3.6, 3.3, 3.5, and 3.4 seconds. Test at the 5% level of significance whether Program A gives a lower mean reaction time.

Code-

Data

A <- c(2.8, 3.1, 2.9, 3.0, 3.2, 2.7, 3.0, 2.8)

B <- c(3.4, 3.7, 3.5, 3.8, 3.6, 3.3, 3.5, 3.4)

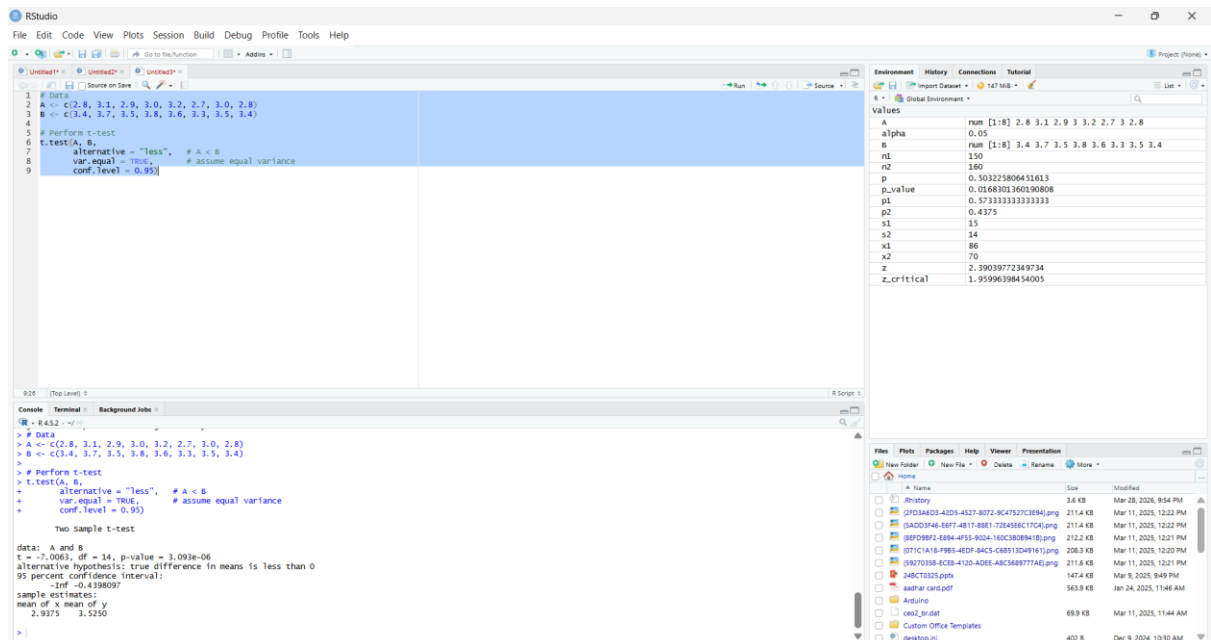
Perform t-test

t.test(A, B,

alternative = "less", # A < B

var.equal = TRUE, # assume equal variance

conf.level = 0.95)



4 A public health researcher wants to know whether mobile phone usage habit is related to sleep quality among college students. A sample of students is classified into three phone-usage groups: low, moderate, and high. Their sleep quality is recorded as good or poor, and the following table is observed: low usage has 28 good and 12 poor sleepers, moderate usage has 24 good and 16 poor sleepers, and high usage has 18 good and 22 poor sleepers. Test whether phone usage habit and sleep quality are independent at the 5% level.

Code-

```
# Create contingency table
```

```
data <- matrix(c(28,12,
                 24,16,
                 18,22),
              nrow = 3,
              byrow = TRUE)
```

```
# Add row/column names (optional)
```

```
rownames(data) <- c("Low", "Moderate", "High")
```

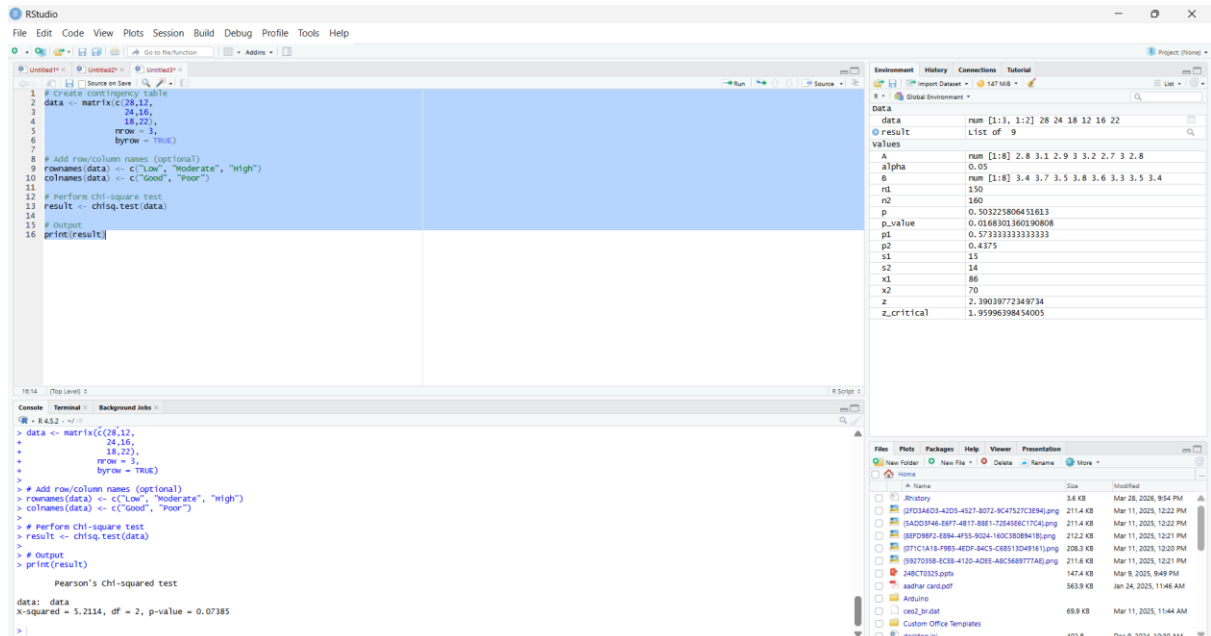
```
colnames(data) <- c("Good", "Poor")
```

```
# Perform Chi-square test
```

```
result <- chisq.test(data)
```

```
# Output
```

```
print(result)
```



- 5 A food brand claims that its snack packs are filled in the proportions 20%, 20%, 20%, 20%, and 20% across five flavor categories. In a quality-control check, 100 packets are inspected and the observed counts are 18, 22, 19, 21, and 20. Test whether the observed distribution agrees with the claimed distribution at the 5% level.

Code-

```
# Observed frequencies
```

```
observed <- c(18, 22, 19, 21, 20)
```

```
# Expected proportions
```

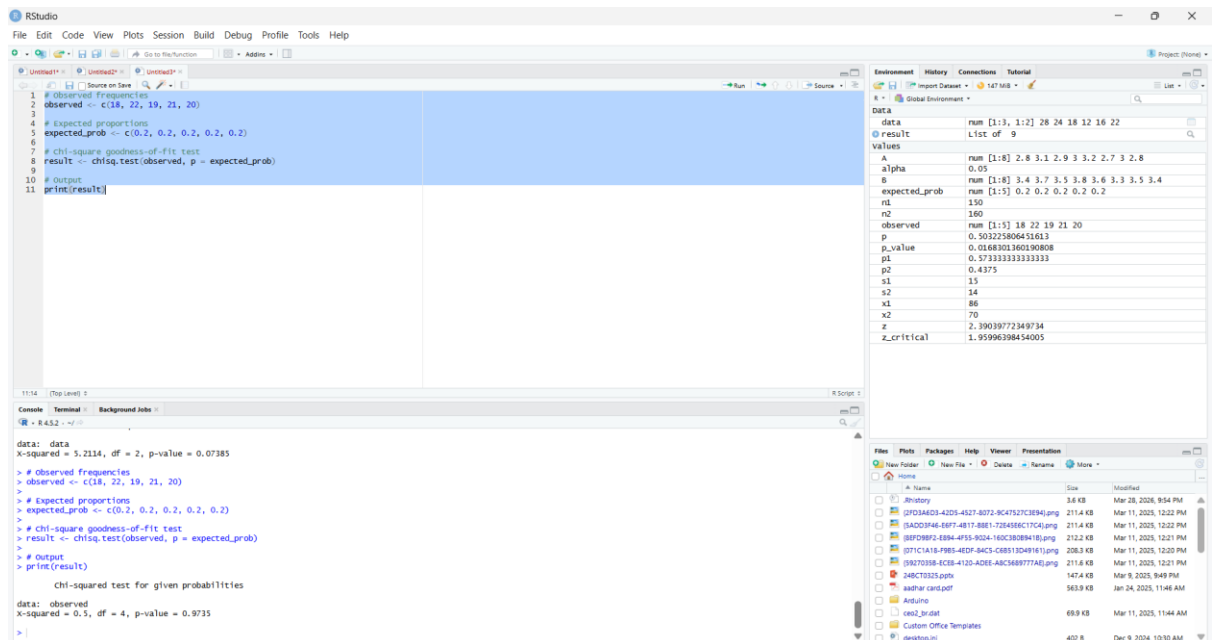
```
expected_prob <- c(0.2, 0.2, 0.2, 0.2, 0.2)
```

```
# Chi-square goodness-of-fit test
```

```
result <- chisq.test(observed, p = expected_prob)
```

```
# Output
```

```
print(result)
```



- 6 A biotechnology lab studies the number of successful gene edits in 10 trials for each sample. The lab believes the number of successes follows a binomial distribution with $n = 10$ and success probability $p = 0.4$. A sample of 200 experiments gives the following observed frequencies for 0 to 10 successes: 5, 14, 23, 31, 36, 34, 25, 16, 9, 5, and 2. Test whether a binomial distribution with $n = 10$, $p = 0.4$ fits the data well.

Code-

Observed frequencies

```
observed <- c(5,14,23,31,36,34,25,16,9,5,2)
```

Parameters

```
n <- 10
```

```
p <- 0.4
```

```
N <- 200
```

Possible values (0 to 10)

```
x <- 0:10
```

Expected probabilities using binomial

```
prob <- dbinom(x, size = n, prob = p)
```

Expected frequencies

```
expected <- N * prob
```

Combine small expected frequencies if needed

(Chi-square requires expected ≥ 5 ideally)

```
observed_adj <- c(observed[1:8], sum(observed[9:11]))
```

```
expected_adj <- c(expected[1:8], sum(expected[9:11]))
```

Chi-square test

```
chisq <- sum((observed_adj - expected_adj)^2 / expected_adj)
```

Degrees of freedom

```
df <- length(observed_adj) - 1 - 0 # no parameter estimated
```

p-value

```
p_value <- 1 - pchisq(chisq, df)
```

Output

```
cat("Chi-square =", chisq, "\n")
```

```
cat("Degrees of freedom =", df, "\n")
```

```
cat("p-value =", p_value, "\n")
```

Decision

```
if(p_value < 0.05){
```

```
  cat("Reject H0: Not a good fit\n")
```

```
} else {
```

```
  cat("Fail to reject H0: Good fit\n")
```

```
}
```

```
12 # Expected probabilities using binomial
13 prob <- dbinom(x, size = n, prob = p)
14
15 # Expected frequencies
16 expected <- N * prob
17
18 # Combine small expected frequencies if needed
19 # (Chi-square requires expected >= 5 ideally)
20 observed_adj <- c(observed[1:8], sum(observed[9:11]))
21 expected_adj <- c(expected[1:8], sum(expected[9:11]))
22
23 # Chi-square test
24 chisq <- sum((observed_adj - expected_adj)^2 / expected_adj)
25
26 # Degrees of freedom
27 df <- length(observed_adj) - 1 - 0 # no parameter estimated
28
29 # p-value
30 p_value <- 1 - pchisq(chisq, df)
31
32 # Output
33 cat("Chi-square =", chisq, "\n")
34 cat("Degrees of freedom =", df, "\n")
35 cat("p-value =", p_value, "\n")
36
37 # Decision
38 if(p_value < 0.05){
39   cat("Reject H0: Not a good fit\n")
40 } else {
41   cat("Fail to reject H0: Good fit\n")
42 }
```

Console

```
> # Degrees of freedom
> df <- length(observed_adj) - 1 - 0 # no parameter estimated
>
> # p-value
> p_value <- 1 - pchisq(chisq, df)
>
> # Output
> cat("Chi-square =", chisq, "\n")
Chi-square = 106.1303
> cat("Degrees of freedom =", df, "\n")
degrees of freedom = 8
> cat("p-value =", p_value, "\n")
p-value = 0
>
> # Decision
> if(p_value < 0.05){
+   cat("Reject H0: not a good fit\n")
+ } else {
+   cat("Fail to reject H0: good fit\n")
+ }
Reject H0: not a good fit
```

Environment

Object	Class	Attributes
data	num	[1:3, 1:2] 28 24 18 12 16 22
result	List of 9	
values		
A	num	[1:8] 2.8 3.1 2.9 3.2 2.7 3.2 8
alpha	num	0.05
B	num	[1:8] 3.4 3.7 3.5 3.8 3.6 3.3 3.5 3.4
chisq	num	106.130200649353
df	num	8
expected	num	[1:11] 1.21 8.06 24.19 43 50.16 ...
expected_adj	num	[1:9] 1.21 8.06 24.19 43 50.16 ...
expected_prob	num	[1:5] 0.2 0.2 0.2 0.2 0.2
n	num	10
nt	num	200
n2	num	150
observed	num	[1:11] 5 14 23 31 36 34 25 16 9 5 ...
observed_adj	num	[1:9] 5 14 23 31 36 34 25 16 16
p	num	0.4
p_value	num	0
p1	num	0.573333333333333
p2	num	0.4375
prob	num	[1:11] 0.00605 0.04031 0.12093 0.21499 0.2 ...
s1	num	15
s2	num	14
x	num	[1:11] 0 1 2 3 4 5 6 7 8 9 ...
x1	num	86
x2	num	70
z	num	2.39039772349734
z2	num	4.0606308454006

Files

Name	Size	Modified
.Rhistory	3.6 KB	Mar 28, 2025, 9:54 PM
[2FD3A6D3-42D5-4527-8072-8C47927C894].png	211.4 KB	Mar 11, 2025, 12:22 PM
[5AD29F46-89F7-4917-8881-7204086C7C04].png	211.4 KB	Mar 11, 2025, 12:22 PM
[8F0F9F02-6284-4F55-8024-16023089416].png	212.2 KB	Mar 11, 2025, 12:21 PM
[D71C1A18-F985-4EDF-84C5-C88513D49F61].png	208.8 KB	Mar 11, 2025, 12:20 PM
[59270358-4CE8-4120-AD85-ABC5688777A].png	211.6 KB	Mar 11, 2025, 12:21 PM
248CT0325.pdf	147.4 KB	Mar 9, 2025, 9:49 PM
asuntra-cans.pdf	563.9 KB	Jan 24, 2025, 11:48 AM
Asuntra		
ceo2_data	69.8 KB	Mar 11, 2025, 11:44 AM
Custom Office Templates		
desktop.ini	402 B	Dec 9, 2024, 10:30 AM

- 7 A call center records the number of customer complaints arriving per hour. The manager believes the number of complaints follows a Poisson distribution with mean $\lambda = 2$. Over 150 hours, the observed frequencies of 0, 1, 2, 3, 4, and 5 or more complaints are 18, 34, 41, 30, 17, and 10. Test whether the Poisson model fits the data at the 5% level.

Code-

Observed frequencies

```

observed <- c(18, 34, 41, 30, 17, 10)

# Parameters
lambda <- 2
N <- 150

# Values (0 to 4)
x <- 0:4

# Poisson probabilities
prob <- dpois(x, lambda)

# Expected frequencies for 0-4
expected <- N * prob

# Expected for 5 or more
expected_5plus <- N * (1 - sum(prob))

# Combine expected
expected <- c(expected, expected_5plus)

# Chi-square statistic
chisq <- sum((observed - expected)^2 / expected)

# Degrees of freedom
# df = categories - 1 - parameters estimated
df <- length(observed) - 1 - 0 # lambda given

# p-value
p_value <- 1 - pchisq(chisq, df)

# Output
cat("Chi-square =", chisq, "\n")
cat("Degrees of freedom =", df, "\n")
cat("p-value =", p_value, "\n")

# Decision
if(p_value < 0.05){
  cat("Reject H0: Poisson is not a good fit\n")
} else {
  cat("Fail to reject H0: Poisson fits well\n")
}

```

RStudio

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Source

```
13 # Expected frequencies for 0-4
14 expected <- N * prob
15 # Expected for 5 or more
16 expected_splus <- N * (1 - sum(prob))
17 # combine expected
18 expected <- c(expected, expected_splus)
19 # Chi-square statistic
20 chisq <- sum((observed - expected)^2 / expected)
21 # degrees of freedom
22 df <- categories - 1 - parameters_estimated
23 df <- length(observed) - 1 - 0 # lambda given
24 # p-value
25 p_value <- 1 - pchisq(chisq, df)
26 # Output
27 cat("Chi-square =", chisq, "\n")
28 cat("degrees of freedom =", df, "\n")
29 cat("p-value =", p_value, "\n")
30 # Decision
31 if(p_value < 0.05){
32   cat("Reject H0: Poisson is not a good fit\n")
33 } else {
34   cat("Fail to reject H0: Poisson fits well\n")
35 }
36
```

Console

```
R 4.4.2 > # df = categories - 1 - parameters estimated
R 4.4.2 > df <- length(observed) - 1 - 0 # lambda given
R 4.4.2 >
R 4.4.2 > # p-value
R 4.4.2 > p_value <- 1 - pchisq(chisq, df)
R 4.4.2 >
R 4.4.2 > # Output
R 4.4.2 > cat("Chi-square =", chisq, "\n")
R 4.4.2 > cat("degrees of freedom =", df, "\n")
R 4.4.2 > cat("p-value =", p_value, "\n")
R 4.4.2 > # Decision
R 4.4.2 > if(p_value < 0.05){
R 4.4.2 +   cat("Reject H0: Poisson is not a good fit\n")
R 4.4.2 + } else {
R 4.4.2 +   cat("Fail to reject H0: Poisson fits well\n")
R 4.4.2 + }
R 4.4.2 >
R 4.4.2 > Fail to reject H0: Poisson fits well
R 4.4.2 >
```

Environment

Data

data	num [1:3, 1:2]	28	24	18	12	16	22
alpha	num [1:8]	2.8	3.1	2.9	3.2	2.7	3.2
B	num [1:8]	3.4	3.7	3.5	3.8	3.3	3.5
chisq	num [1:1]	3.1028355371943					
df	num [1:1]	5					
expected	num [1:6]	20.3	40.6	40.6	27.1	13.5	...
expected_splus	num [1:1]	7.89795260155666					
expected_adj	num [1:9]	1.21	8.06	24.19	43	50.16	...
expected_prob	num [1:5]	0.2	0.2	0.2	0.2	0.2	
lambda	num [1:1]	2					
n	num [1:1]	10					
n1	num [1:1]	150					
n2	num [1:1]	160					
observed	num [1:6]	18	34	41	30	17	10
observed_adj	num [1:9]	5	14	23	31	36	25
p	num [1:1]	0.4					
p_value	num [1:1]	0.684135200582603					
p1	num [1:1]	0.573333333333333					
p2	num [1:1]	0.4375					
prob	num [1:5]	0.1353	0.2707	0.2707	0.1804	0.0902	
s1	num [1:1]	15					
s2	num [1:1]	14					
x	num [1:5]	0	1	2	3	4	
x1	num [1:1]	86					
x2	num [1:1]	76					

Files

Name	Size	Modified
History	3.6 KB	Mar 28, 2025, 9:54 PM
2FD3A6D5-43D5-4527-8C47527C3E94.png	211.4 KB	Mar 11, 2025, 12:22 PM
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871C1A18-F985-4EDF-8AC5-C68513D49F16.png	208.3 KB	Mar 11, 2025, 12:20 PM
59270358-ECB8-412D-AD8E-ABC5689777AE.png	211.6 KB	Mar 11, 2025, 12:21 PM
248C70325.png	147.4 KB	Mar 9, 2025, 9:49 PM
aadhar card.pdf	563.9 KB	Jan 24, 2025, 11:46 AM
Aadhar		
ee62_2r.dat	68.9 KB	Mar 11, 2025, 11:44 AM
Custom Office Templates		
desktop.ini	402 B	Dec 9, 2024, 10:30 AM